Dataset and Benchmark for Fine-Grained Ship Recognition in Complex Optical Remote Sensing Scenarios

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***Abstract*— Ship recognition in remote sensing imagery is crucial for numerous applications such as monitoring maritime security, preventing illegal activities, and implementing environmental protection. However, most existing research rarely focuses on both the complexity of the scene and the fine-grained recognition of targets, which limits the accuracy and applicability of ship recognition technology. To propel advancements in ship recognition methodology, we propose a new dataset named fine-grained ship recognition in complex scenarios (FGSRCSs), which contains 17 typical categories of large and medium-sized ships from 280 widely distributed ports. For the richness of the image characteristics, the dataset images are collected from multiple satellite platforms, such as Ikonos, Jilin-1, OrbView, Pleiades, and WorldView, and the time span of the data covers the past 20 years. To ensure the diversity of scenarios, the dataset mainly considers five complex scenarios, including thick clouds, mists, shadows, sea clutter, and ground facilities, which helps to train and improve the algorithm's applicability to practical application scenarios. Furthermore, we conduct experiments with ten state-of-the-art recognition algorithms on the FGSRCS dataset, providing a benchmark for algorithm application. The research result can furnish both theoretical insights and practical guidance for the development of future ship recognition models. The dataset is available on https://github.com/dwddw/FGSRCS**

***Index Terms*— Complex scenario, fine-grained ship recognition, optical dataset, performance benchmark, remote sensing image.**

I. Introduction

SHIPS are the primary facilitators of maritime activities, essential for numerous military and civilian applications such as goods transportation, resource exploitation, and emergency rescues at sea [1]. Accurate ship identification technology can significantly improve the reliability and efficiency of maritime applications. Maritime remote sensing systems predominantly rely on three imaging modalities: visible light, infrared, and synthetic aperture radar (SAR). Although SAR and infrared sensors provide all-weather capabilities, their fundamental limitations in spatial resolution and thermal ambiguity restrict fine-grained feature extraction. In contrast, optical remote sensing images have become the predominant data source for ship detection and recognition, owing to their high spatial resolution (submeter precision), visually interpretable RGB spectral characteristics, rich textural details, and widespread availability through commercial satellite constellations and aerial platforms. Particularly, with the continuous improvement in the resolution of remote sensing images and the feature extraction capabilities of deep learning networks, fine-grained recognition of ships based on high-resolution remote sensing images has emerged as a hot research topic in the field of ship image interpretation.

Learning-based methods have shown significant advantages in ship recognition research, yet the performance of deep networks remains critically dependent on the volume and quality of the datasets. A comprehensive ship recognition dataset needs to offer features from multiple dimensions, such as target scale, type, and state, thus facilitating effective learning and generalization during model training. Moreover, the dataset needs to cover a wide range of scenarios and anomalies as much as possible to enhance model robustness and predictive accuracy. Actually, researches have compiled various datasets for ship recognition, such as HRSC2016 and ShipRSImageNet. However, these datasets still face the following challenges.

1) Existing datasets focus on identifying conventional vessels such as aircraft carriers and destroyers, neglecting the most common and diverse target types like littoral combat ships, which restricts the generality of the target type.
2) The images in existing datasets are usually captured under low-interference conditions, with little consideration for complex environmental factors such as thick clouds, shadows, and ground facilities, which limits the algorithms' adaptability in practical applications.
3) Existing datasets mainly rely on some single imaging platforms and collect imagery from a few large ports, resulting in insufficient data on how target characteristics vary under different spatial, temporal, and platform degradation conditions.

To mitigate the above issues, we propose a ship recognition dataset named fine-grained ship recognition in complex scenarios (FGSRCSs). It covers most common types of large and medium-sized ships around the world, including aircraft carrier, amphibious assault ship, amphibious transport dock,

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corvette, cruiser, destroyer, frigate, landing ship, littoral combat ship, mine warfare vessel, patrol vessel, replenishment oiler, submarine, container ship, cruise ship, oil tanker and sand carrier, ensuring the diversity of ship targets. Moreover, the FGSRCS dataset encompasses rich and typical environmental interferences, such as thick clouds, mists, shadows, sea clutter, ground facilities, and some other rare factors. To enhance the diversity of the dataset, we collect data from multiple satellite platforms, covering over 280 famous ports around the world in space and spanning over 20 years in time. In addition, we conduct performance comparison experiments via ten state-of-the-art ship recognition methods, providing a benchmark for subsequent algorithm application.

The contributions of this work can be summarized as follows.

1) The FGSRCS dataset includes 13 typical military ship types and four civilian ship types, covering most categories of the large and medium-sized vessels in the world, which significantly enriches the diversity of the targets within the dataset.
2) The dataset covers a variety of complex environmental interferences, such as thick clouds, mists, shadows, sea clutter, ground facilities, and some other abnormal conditions, which is beneficial for improving the recognition algorithm's applicability to real-world conditions.
3) The FGSRCS dataset comprises multiplatform images spanning of more than 20 years and includes 17 categories sourced from over 280 ports across 58 countries and regions, which ensures the richness of the scene in the spatiotemporal dimension.
4) Performance comparison experiments of multiple state-of-the-art algorithms on the proposed dataset are conducted, establishing a benchmark for FGSRCSs.

II. Related Works

A. Ship Recognition Datasets

Deep learning methods have achieved impressive results in the field of ship recognition. However, the success of deep learning heavily relies on large amounts of accurately annotated data, giving rise to the emergence of numerous ship recognition datasets. Those datasets generally fall into two categories: ship-exclusive datasets and mixed datasets.

1) Ship-Exclusive Datasets: In 2016, the HRSC2016 dataset was constructed [2]. It utilizes publicly available high-resolution aerial imagery and contains 1061 annotated images. The dataset is structured with three levels of instance annotations, encompassing totally 25 categories. However, despite the relatively large number of categories, the dataset contains only around 1000 images, which limits its ability to support deep learning models for advanced recognition tasks due to the insufficient quantity of data.

In 2020, the FGSD dataset collected aerial images from 17 ports across four countries [3]. In addition to various ship categories, the "dock" category has also been introduced, bringing the total number of categories to 43. This dataset also employs multilevel annotations similar to those used in the HRSC2016 dataset. Despite the wide range of categories, the dataset contains only around 5000 instances, which limits its ability to provide sufficient coverage of the multidimensional features of the targets.

In 2021, to further enrich the diversity of data, the ShipRSImageNet dataset was constructed [4]. It broadly collects images from xView, HRSC2016, FGSD, Airbus Ship Detection Challenge, and real-world imagery captured by Chinese satellites such as GaoFen-2 and JiLin-1. This dataset provides detailed four-level annotations for ship types, and at its most granular level, specific ship classes are annotated, such as Arleigh Burke DD, Perry FF, and so on.

In 2022, the SDS dataset was built for ship detection and segmentation [5]. The images are collected from various locations, such as harbors, shorelines, nearby waters, and vase oceans. Due to the scarcity of aircraft carriers in real-world scenarios, SDS adopts a kind of crop-and-paste strategy to generate 395 aircraft carrier training samples, which alleviates the class imbalance issue. In the same year, the VHRShips dataset was introduced [6]. It features 43 cm high-resolution imagery sourced from Google Earth. Images are captured from an eye altitude of 500 m, with those of differing resolutions adjusted through upscaling or downscaling for consistency. Overall, the dataset includes 5312 images containing one or more ships, and 1000 images without ships.

In 2023, the MCSD dataset was released [7]. It is a high-resolution optical multicategory ship detection dataset derived from the GaoFen-2 satellite and Google Earth. It covers 38 global harbors across 16 countries. The ships in the dataset are categorized into 16 types, primarily consisting of civilian vessels.

In 2024, the UOW-Vessel dataset is the latest proposed dataset for ship recognition, and closely resembles the one presented in this article [8]. This dataset includes 3500 images from 14 countries, sourced from historical Google Maps imagery between July 2007 and May 2023. All instances are categorized into ten classes: aircraft carrier, landing, cruiser, destroyer, frigate, patrol, corvette, submarine, other military, and civilian, with annotations provided as polygons and precise segmentation masks. In comparison, our dataset offers more detailed category annotations with 17 classes. In addition, UOW-Vessel is primarily composed of civilian vessels, which make up 68.56% of the total, while our dataset focuses mainly on military ships.

2) Mixed Datasets: Beyond datasets that focus solely on ships, there are also classic ones that include both ships and other types of targets. In 2014, the NWPU VHR'10 dataset was introduced with 800 high-resolution aerial images [9]. It includes 650 positive images with detectable objects and 150 negative images without targets. The dataset features a mix of color images from Google Earth and color infrared images from Vaihingen. The targets are divided into ten categories, including ships. In 2018, the DOTA dataset became a prominent resource extensively utilized for both evaluation and development of object detection algorithms in aerial imagery [10]. The dataset combines high-resolution images acquired from multiple platforms, including Google Earth, GF-2 satellite, and JL-1 satellite, offering a diverse range of perspectives and details. The dataset encompasses 54 353

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TABLE I
STATISTICAL RESULTS OF EXISTING SHIP RECOGNITION DATASETS

Dataset	Year	Data Sources	Images	Ship Instances	Ship classes	Resolution	Annotation
HRSC2016	2016	Google Earth	1061	2886	25	0.4m-2m	HBB, OBB
FGSD	2020	Google Earth	2612	5634	43	0.12m-1.93m	HBB, OBB
ShipRSImageNet	2021	xView, HRSC2016, FGSD, Airbus Ship Detection Challenge, GaoFen-2	3435	17573	50	0.12m-6m	HBB, OBB
SDS	2022	Google Earth	1984	13487	5	-	OBB
VHRShips	2022	Google Earth	6312	11179	34	0.43m	HBB
MCSD	2023	GaoFen-2, Google Earth	4000	18000	16	0.25m-2m	OBB
UOW Vessel	2024	Google Earth	3500	35598	10	-	Polygon
NWPU VHR-10	2014	Google Earth	800	302/3775	1/10	0.5m-2m	HBB
DOTA	2018	Google Earth, GaoFen -2, Jilin-1	5215	54353/349675	1/18	-	HBB, OBB
xView	2018	WorldView-3	1127	4452/601726	9/60	0.3m	HBB
DIOR	2020	Google Earth	23463	62400/192472	1/20	0.5m-30m	HBB
Ours	2024	Google Earth, IKONOS, Jilin-1, OrbView, Pleiades, WorldView	4500	11187	17	0.5m	HBB, OBB

instances of ships. In the same year, the xView dataset was released using imagery from WorldView-3 satellites and encompasses eight major categories [11]. Within the maritime vessel category, there are nine subclasses of ships. In 2020, the DIOR dataset offered a high-resolution benchmark for object detection in optical remote sensing, with 23 463 images and 192 472 instances, including 62 400 ships [12]. Table I provides a detailed summary about existing datasets. For mixed datasets, in the ship classes column, the numbers before the slash represent the count of ship categories, while the numbers after the slash denote the total number of categories.

While existing datasets encompass a variety of ship types, they still neglect some important ship categories such as littoral combat ships and mine warfare vessels. In addition, most datasets rely on a single image source, such as Google Earth, which limits the diversity of images featuring different levels of feature degradation. To overcome these deficiencies, the proposed FGSRCS dataset encompasses 17 common ship categories, providing comprehensive global coverage. Moreover, the dataset also incorporates images from different platforms as well as historical imagery, effectively ensuring the diversity of scenarios. This broadening of scope significantly augments the dataset's comprehensiveness and variety, which can facilitate the development of more sophisticated and robust recognition models.

B. Ship Recognition Methods

Traditional ship recognition methods primarily operate through image preprocessing followed by multidimensional feature extraction [13] to achieve granular interpretation of target information, including those based on gray-scale features [14], template matching [15], visual saliency [16], texture characteristics [17], and machine learning [18], are employed to locate and classify ship instances. However, those methods require prior knowledge and handcrafted features, and their performance often degrades in complex scenarios.

Driven by the evolution of computational techniques, deep learning-based methods have demonstrated remarkable efficacy in critical computer vision tasks, including but not limited to target feature extraction, feature enhancement [19], [20], [21], [22], feature fusion [23], and object recognition [24], [25], substantiating their superior performance in practical applications. In the field of object recognition, deep learning-based methods can be divided into two-stage methods and single-stage methods [26], [27]. The RCNN series algorithms are prominent examples of two-stage methods [28], [29], [30]. They start by generating region proposals using selective search or a region proposal network (RPN). Each proposal is then processed to extract features using a convolution neural network (CNN). These extracted features are subsequently classified to identify the object class and refined to adjust the bounding box accurately. Single-stage methods, such as RetinaNet [31], YOLO [32], and FCOS [33], streamline the process by directly predicting the class and bounding box without generating region proposals or performing non-maximum suppression (NMS) operations. DETR is a novel single-stage detector that employs the powerful self-attention mechanism inherent in Transformer to process the entire image as a sequence of patches, allowing for a more comprehensive and contextual understanding of the scene [34]. DETR simplifies the process of object detection by eliminating the need for many of the heuristic and hand-engineered components, such as anchor generation and NMS. While two-stage methods generally achieve higher accuracy, single-stage methods are faster and more efficient.

With the development of ship recognition datasets, a range of methodologies for detecting and recognizing ship targets have been progressively introduced. In 2020, CR2A-Net proposed a cascading network structure for ship detection [35]. The model first segments the input images into smaller slices, then uses a binary classifier to determine whether ships are present in these slices. Slices confirmed to contain ships are subject to further processing, where the rotated align convolution (RAC) module refines image features to enhance the accuracy of ship detection. In 2021, ReDet [36] devised a rotation-equivariant backbone and rotation-invariant RoI Align, achieving both rotation-equivariant and rotation-invariant feature extraction. In 2022, DSLA designed a novel anchor quality score that incorporates both prior and predictive data of the anchor, allowing the model to actively engage in the label assignment process [37]. In addition, a soft label assignment strategy was developed to assign weights to training samples within the loss function, effectively reducing inconsistencies between the regression and classification outputs. In 2024, SASOD integrated a saliency-guided feature fusion network (SGFFN) and a dynamic IoU-adaptative strategy (DIAS) to address common issues such as inaccurate

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localization of ships against cluttered backgrounds and the detection of small ships [38]. The SGFFN comprises a resolution-matching saliency supervision (RMS) module and a cross-stage saliency integration network (CSIN) to refine and merge saliency-aware features for ship visibility. Meanwhile, the DIAS customizes training for small ship detection by adjusting IoU thresholds. These ship recognition methods have achieved good results on datasets with relatively clean scenes, but there is a significant decrease in performance under complex scene interference conditions.

For ship recognition in complex scenarios, scholars have proposed innovative approaches to address challenges such as background clutter, multiscale targets, and dense distributions. In 2022, EIRNet adopts a dense feature fusion network (DFF-Net) and a dual-mask attention module (DMAM), optimizing multiscale feature reuse and clutter suppression to refine localization and fine-grained recognition for densely distributed ships [39]. In 2023, KCPNet employs a balanced feature fusion network (BFF-Net) and a contextual attention network (CA-Net), leveraging knowledge-driven optimization to improve robustness against small targets and mitigate clutter interference [40]. In 2024, SMPISD-MTPNet integrates a scene semantic extractor (SSE) to extract the scene's semantic information, effectively suppressing false alarms in cluttered maritime environments while enhancing recognition accuracy for small and dim targets [41].

The success of the aforementioned algorithms heavily relies on the support of datasets, such as DOTA, HRSC2016. However, these datasets typically feature clean backgrounds and fail to adequately account for complex scenarios such as clouds and shadows, thereby constraining the robustness of the algorithms. Different from other datasets, the FGSRCS dataset encompasses a range of complex environmental factors such as thick clouds, mists, shadows, sea clutter, ground facilities, and rare conditions, which enhances the capability of recognition algorithms to perform high-precision and robust recognition in such difficult scenarios.

III. Dataset Construction

A. Construction Method

To provide a comprehensive range of scenarios, we capture images from more than 280 ports, naval bases, and shipyards across 58 countries and regions. Typical countries and major ports are illustrated in Table II. The collection location of the dataset images is shown in Fig. 1. To meet the requirements of fine-grained ship recognition tasks, the captured images have a ground resolution of approximately 0.5–1 m. To further increase the diversity of the dataset, we collect historical imagery from 2004 to 2024, which ensures the inclusion of a richer variety of viewing angles, weather conditions, and background changes. Translation and rotation strategies are employed during the image processing procedure. To accommodate practical applications, we gather imagery from the Ikonos, Jilin-1, OrbView, Pleiades, and WorldView satellites, with the number of images being 12, 32, 7, 9, and 87, respectively. Compared to Google Earth, these images do not contain preprocessing or manual selection, making them closer to natural imaging scenarios. In total, we collected 4500 images and 15 666 ship instances.

TABLE II
Examples of Naval and Commercial Port Cities Worldwide

Country	City
USA	Bremerton, Norfolk, San Diego, Philadelphia, Jacksonville
UK	Plymouth, Portsmouth, Liverpool, Rosyth
India	Mumbai, Karwar, Kochi, Visakhapatnam
France	Toulon, Brest
Germany	Wilhelmshaven, Hamburg, Kiel
Italy	La Spezia, Taranto, Augusta
Japan	Sasebo, Kure, Maizuru, Yokosuka
Australia	Perth, Sydney
Russia	Vilkovo, Murmansk, Severomorsk

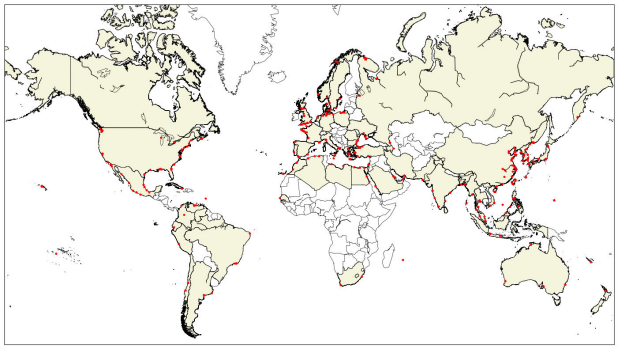


Fig. 1. Diagram of port location distribution in dataset images.

As far as we know, there is no unified standard for the taxonomy of ships in typical datasets, particularly for military vessels. Datasets such as HRSC2016, FGSD, and UOW-Vessel each use different criteria. We devoted significant effort to gathering information on ships from multiple countries, including top-down views, length-width data, and the function of each ship, to assist in data annotation. Considering these factors and the data downloaded, we remove ship categories with fewer samples and consolidate some categories, resulting in a final set of 17 categories. For example, as replenishment oiler and fast combat support ship have similar functions and appearance, they were grouped into replenishment oiler, while expedition transfer dock and hospital ship were removed due to a lack of sufficient samples. Table III gives the detailed information of typical category allocation.

All annotations are made using rotated boxes, which are widely used in aerial object detection tasks and can introduce less background noise. For images under complex scenarios, we assign them a difficulty level of 1. The final annotation results are presented in the DOTA format, namely, $x1$, $y1$, $x2$, $y2$, $x3$, $y3$, $x4$, $y4$, category, and difficulty.

In addition, to accommodate various practical needs, we provide multilevel annotations for the categories mentioned above. Since some ships do not clearly belong to a specific category, so in Level 0, all ship targets are labeled as "ship," which can be used for detection tasks. In level 1, ships are divided into "military" and "civilian." Level 2 offers the finest granularity of annotation. For military vessels, we have aircraft carrier, amphibious assault ship, amphibious transport

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TABLE III
CRITERIA FOR THE CLASSIFICATION OF SEVERAL SHIP CATEGORIES

Category	Description
Aircraft Carrier	including helicopter carrier, as they are both designed for launching and recovering aircraft
Cruiser	including battlecruiser, as they are similar in displacement, armament and cost
Frigate	including destroyer escort, as the US navy identifies destroyer escort as a frigate
Landing Ship	including landing ship tank, landing ship medium, landing ship dock and landing ship craft, as they are all designed to transport troops or vehicles for amphibious assaults
Mine Warfare Vessel	including mine countermeasures vessel, minehunter, minelayer and minesweeper, as they all specialize in missions related to mines
Patrol Vessel	including gunboat, missile boat, fast attack boat, submarine chaser, torpedo boat and coast guard vessel, as they are all agile boats used for coastal defense and patrolling
Replenishment Oiler	including fast combat support ship, as they both serve to provide logistical support to naval vessels at sea and share similar appearances
Sand Carrier	including bulker, as they are both civilian ships used to transport solid goods and have similar structural designs

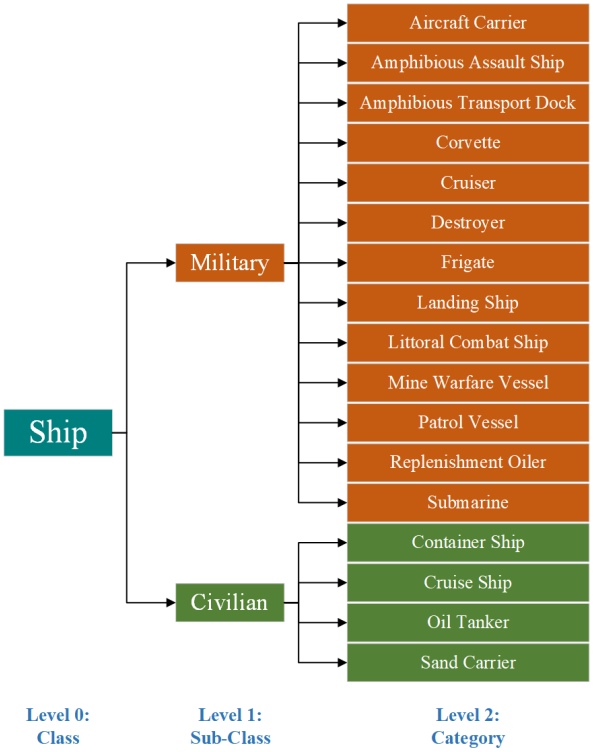


Fig. 2. FGSRCS class hierarchy.

dock, corvette, cruiser, destroyer, frigate, landing ship, littoral combat ship, mine warfare vessel, patrol vessel, replenishment oiler, and submarine. For civilian vessels, we have container ship, cruise ship, oil tanker, and sand carrier. Note that the category "ship" only exists in Level 0 for the detection task and is not presented in Level 1 and Level 2 annotations. With the taxonomy described above, we ensure the coverage of most large and medium-sized ship types, while avoiding overlapping labels. The class hierarchy of the FGSRCS dataset is shown in Fig. 2. Moreover, example images of a typical ship in the FGSRCS dataset are shown in Fig. 3.

B. Dataset Properties

The FGSRCS dataset comprises 4500 images, containing 11 187 ship instances categorized into 17 distinct classes. As detailed above, ships that do not belong to any specific category are also annotated. Each image has a resolution of 1024 × 1024 pixels, with an approximate ground resolution of 0.5 m.

The co-occurrence matrix of different of ship categories is shown in Fig. 4. The diagonal values represent the counts of instances for each specific ship category, whereas the off-diagonal values quantify the frequency with which pairs of different categories co-occur within the same image. Fig. 4 illustrates that the FF category has the highest number of instances in the dataset, totally 2594 instances. In contrast, the LCS category is the least represented, with 158 instances. In addition, the analysis suggests a pronounced tendency for certain warship categories, such as FF and PV, to appear concurrently in the same images.

The FGSRCS dataset comprises 17 categories, with the pixel dimensions of each instance within these categories depicted in Fig. 5. The average aspect ratios for the various categories are presented in the upper left corner of the figure. It is clearly demonstrated that the aspect ratios of all ship instances surpass 4, with an average value of 5.74. Submarine presents the highest aspect ratio at 8.87, while ac records the lowest at 4.20. This indicates that ship targets have significantly higher aspect ratios compared to typical objects. Consequently, OBB detection is necessary to accurately capture their elongated shapes and orientations. It should be noted that the calculated aspect ratios for SUB could be somewhat overstated, as these vessels are partially submerged. The aspect ratios for other categories may be modestly understated due to perspective distortions from oblique viewing angles.

To improve the performance of detection and recognition algorithms under realistic conditions, the FGSRCS dataset has compiled more than 500 images captured in five challenging environments, including thick clouds, mists, shadows, sea clutter, and complex facilities, as well as some rare conditions, such as sea ice and overexposure. Representative samples from these environments are shown in Fig. 6. With the addition of these supplementary images, algorithms trained on this dataset can be better equipped to handle the variability and challenges of real-world settings, thus enhancing resilience and accuracy across diverse operational environments.

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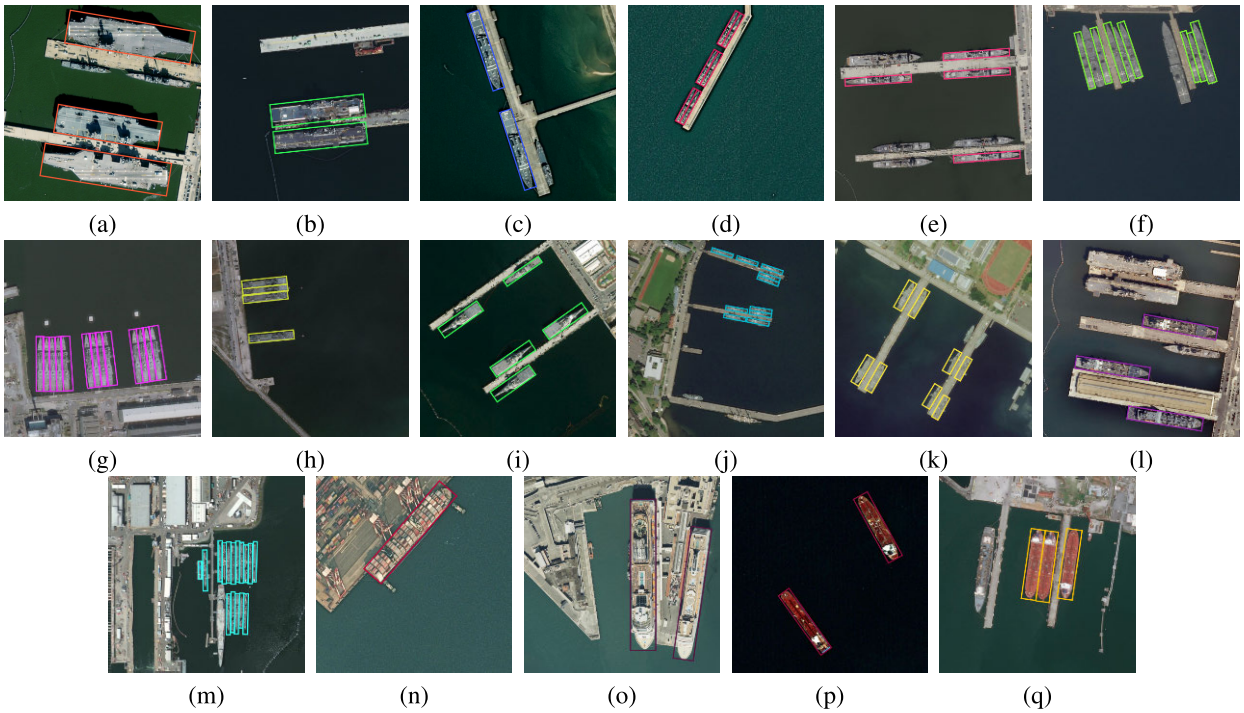


Fig. 3. Ship categories in proposed dataset. The abbreviations for categories are defined as: (a) AC: aircraft carrier. (b) AAS: amphibious assault ship. (c) ATD: amphibious transport dock. (d) CVT: corvette. (e) CR: cruiser. (f) DD: destroyer. (g) FF: frigate. (h) LS: landing ship. (i) LCS: littoral combat ship. (j) MWV: mine warfare vessel. (k) PV: patrol vessel. (l) RO: replenishment oiler. (m) SUB: submarine. (n) CTS: container ship. (o) CRS: cruise ship. (p) OT: oil tanker. (q) SC: sand carrier.

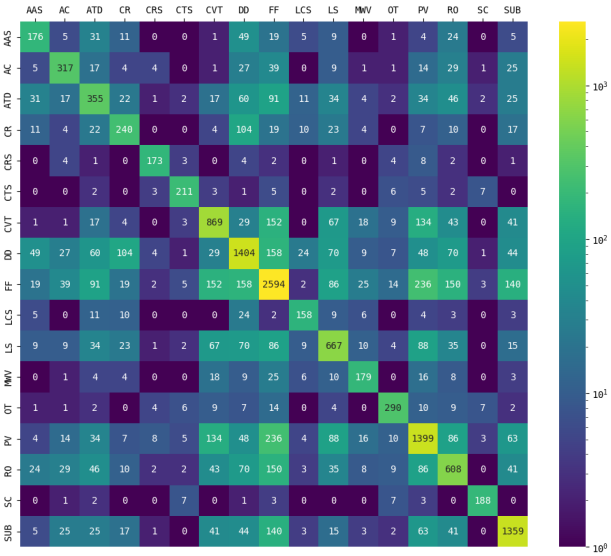


Fig. 4. Co-occurrence matrix of different ship categories.

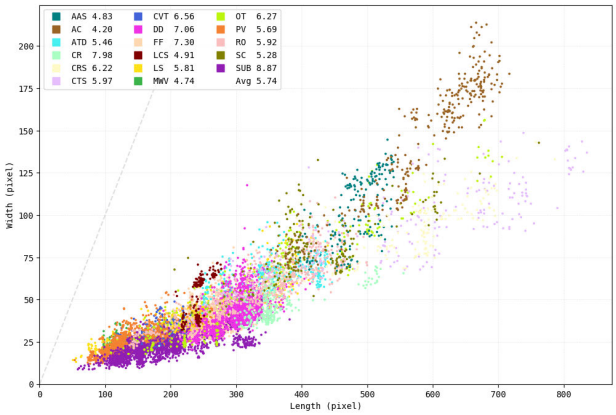


Fig. 5. Scale distribution of ship instances.

IV. Ship Recognition Benchmark

A. Evaluation Metrics

We employ mean average precision (mAP) to ascertain the efficacy of ship recognition algorithms within the newly constructed dataset. Precision (P) is defined as the ratio of accurately identified objects (true positives) to all detected items, inclusive of both correct and mistaken recognitions. Recall measures the model's capacity to identify all relevant items, computed as the proportion of true positives over the sum of true positives and false negatives.

AP provides a robust measure of an algorithm's precision-recall balance by calculating the area under the precision and recall curve. The metric offers a nuanced assessment of recognition performance, capturing both the accuracy and comprehensiveness of the model in identifying relevant objects. For multiclass recognition scenarios, mAP is typically evaluated per class and then averaged to furnish a singular metric reflective of overall performance.

These metrics are defined as follows:

$$P = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}} \tag{1}$$

$$R = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{2}$$

$$\mathrm{AP} = \int_0^1 P(R)\, dR \tag{3}$$

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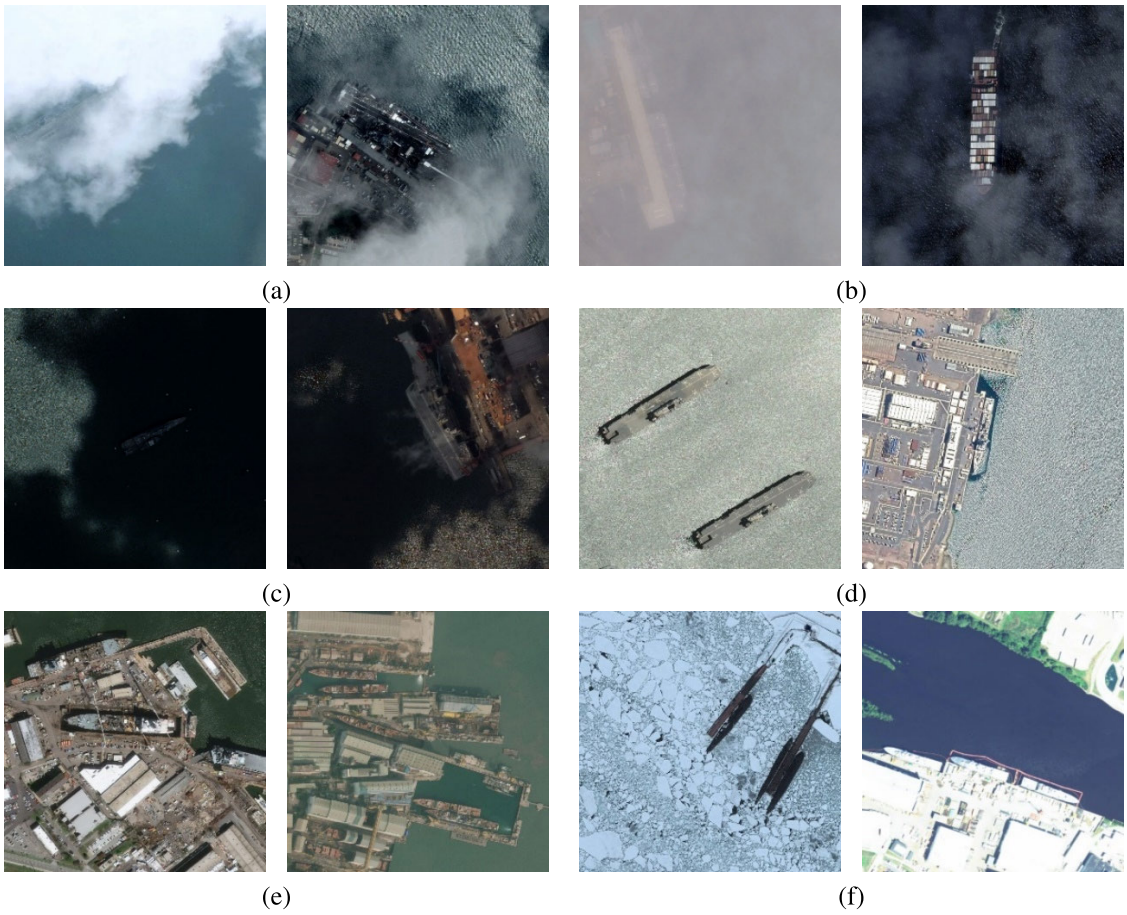


Fig. 6. Complex environmental interferences in proposed dataset. (a) Thick clouds. (b) Mists. (c) Shadows. (d) Sea clutter. (e) Ground facilities. (f) Other rare conditions.

$$\text{mAP} = \sum_{i=1}^{N} \text{AP}_i \tag{4}$$

where TP, FP, and FN denote true positives, false positives, and false negatives, respectively.

B. Implementation Details

To guarantee impartial evaluations, we utilized the well-established MMRotate framework for our directional object recognition experiments. Due to the high interclass similarity among ship categories, we apply NMS collectively across all categories to prevent the generation of overlapping detection boxes for different classes. All training and evaluation were conducted using Level 2 annotations, which are the finest-grained annotations. The experiments were conducted on a high-performance computing system equipped with an Nvidia RTX 4090 GPU. To enable consistent comparisons, each model was trained over 36 epochs with a set batch size of 2. All configuration files used during training were modified from those used for the DOTA dataset, with changes limited to updating the dataset paths.

C. Algorithm Performance Comparison

To comprehensively assess the recognition performance of the proposed dataset, we employ nine well-established oriented object recognizers for comparative analysis. These recognizers include Rotated RetinaNet, S2ANet [42], R3Det [43], Rotated Faster RCNN, EIRNet, ROI Transformer [44], Oriented-RCNN [45], ReDet, SASM Reppoints [46], and ARS DETR [47]. These methods are models widely used in ship recognition research in recent years, and their codes are all open-source.

Specifically, Rotated RetinaNet utilizes focal loss to address the class imbalance issue, while introducing rotating bounding boxes and an optimized anchor strategy that enhances the model's ability to accurately detect and classify objects at various orientations. S2ANet comprises a feature alignment module (FAM) and an orientation detection module (ODM). The FAM refines anchor quality and aligns convolutional features to the anchors for improved accuracy, while ODM uses rotating filters to encode orientation, producing features that enhance the consistency between location and classification. R3Det builds upon the standard single-stage detection framework by incorporating a feature refinement module (FRM), which enhances feature alignment by reencoding positional information of refined bounding boxes, thereby significantly improving detection accuracy. Rotated Faster RCNN introduces RPN to directly generate region proposals from convolutional features, enhancing both speed and accuracy efficiency. EIRNet uses DFF-Net for robust multiscale detection, DMAM for clutter suppression and

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TABLE IV
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT ALGORITHMS

	AC	AAS	ATD	CVT	CR	DD	FF	LS	LCS	MWV	PV	RO	SUB	CTS	CRS	OT	SC	mAP
Single-Stage																		
Rotated RetinaNet	73.9	47.6	38.7	34.7	24.4	47.3	52.8	34.5	47.3	3.0	47.2	44.0	70.6	79.5	59.0	52.7	54.6	47.8
	59.4	35.4	38.4	24.8	16.7	37.8	40.6	25.9	41.2	3.0	32.0	42.9	19.9	57.0	43.6	34.6	47.8	35.4
S2ANet	66.1	65.6	72.6	45.4	40.8	52.6	66.1	45.0	52.7	9.1	51.6	61.0	79.8	81.8	79.8	73.7	69.7	59.6
	50.1	50.3	64.6	36.9	40.8	50.2	63.8	36.6	52.7	9.1	43.1	53.0	32.9	79.2	76.1	68.9	50.1	50.5
R3Det	86.6	76.4	66.3	43.9	58.2	63.1	70.6	48.6	70.9	25.6	56.6	60.6	75.6	89.3	90.4	76.1	80.0	67.0
	78.3	68.1	61.8	32.1	47.8	55.6	55.8	41.5	67.2	25.6	40.6	60.2	22.2	72.0	76.8	55.8	70.2	54.8
Two-Stage																		
Rotated Faster RCNN	79.4	78.0	72.2	50.0	58.3	57.0	59.4	48.5	70.4	28.2	46.1	68.7	60.8	90.4	76.6	68.7	71.0	63.8
	43.4	37.4	51.2	30.0	28.3	41.9	31.9	20.9	16.4	10.0	15.6	46.3	7.4	34.2	24.6	38.0	37.6	30.3
EIRNet	87.9	88.4	83.4	57.9	70.3	68.3	77.1	69.4	80.7	32.5	67.0	79.1	78.4	90.7	90.9	86.5	83.3	76.0
	76.9	72.2	79.2	51.8	64.1	65.0	70.1	60.4	69.8	28.9	55.0	74.6	43.3	84.3	87.7	76.8	72.9	66.6
ROI Transformer	89.3	90.6	79.7	57.6	72.1	69.2	79.5	70.4	81.6	26.1	72.1	81.2	81.2	90.7	90.9	89.8	81.8	76.7
	79.3	78.6	79.7	56.3	72.1	69.1	78.7	70.4	81.4	26.1	64.0	81.2	56.2	90.7	90.9	89.7	81.8	73.3
Oriented RCNN	89.4	89.9	89.4	59.3	71.3	70.0	79.9	77.7	81.6	35.2	70.9	80.8	81.1	90.4	100	90.4	90.5	79.3
	87.4	79.0	89.4	57.0	71.1	69.5	79.1	69.4	81.6	35.2	62.9	80.8	45.4	76.8	90.1	81.7	81.4	72.8
ReDet	90.7	90.9	89.3	65.2	79.3	79.1	88.3	72.2	90.7	48.0	74.1	81.5	89.8	90.9	90.9	90.1	81.6	81.9
	90.7	90.9	89.3	57.2	79.3	78.8	79.5	71.8	90.7	48.0	72.2	81.5	65.0	90.9	90.7	90.1	81.6	79.3
Point-Based																		
SASM Reppoints	74.3	66.9	56.7	42.3	45.7	55.0	70.1	37.7	54.5	3.0	47.7	50.8	67.8	66.3	70.1	50.4	46.4	53.3
	65.9	55.5	55.4	34.3	44.2	51.7	59.4	33.5	54.5	3.0	35.8	50.8	25.9	58.2	62.6	42.5	41.7	45.6
Transformer-Based																		
ARS DETR	86.8	84.1	78.1	53.9	65.2	67.7	75.7	54.8	70.9	32.3	50.9	81.8	65.6	88.5	88.7	70.5	81.5	70.4
	83.3	82.5	75.8	46.1	63.2	65.4	70.9	49.9	70.9	27.0	43.8	80.4	28.9	82.0	88.7	64.7	77.7	64.8

target enhancement, and achieves robust multiscale recognition. ROI Transformer learns the mapping from horizontal ROI to rotated ROI, this transformation addresses the primary challenge of detecting objects with arbitrary orientations in aerial images. Oriented RCNN is composed of Oriented RPN and Oriented RCNN Head. The former efficiently generates high-quality oriented proposals with minimal computational cost, and the latter refines and recognizes oriented ROIs. ReDet introduces rotation-equivariant networks within the detection framework, which enhances the feature extraction process. Furthermore, the novel rotation-invariant ROI align complements this by extracting rotation-invariant features from the rotation-equivariant ones, optimizing both spatial and orientation dimensions. SASM reppoints is a point-based detection framework, which introduces two innovative shape-adaptive strategies: shape-adaptive selection (SA-S) and shape-adaptive measurement (SA-M). The SA-S strategy identifies and selects high-quality candidate samples, considering the distribution of shape and characteristic features of the objects. The SA-M strategy assigns quality weights to these samples, thereby incorporating additional quality information and enhancing the overall detection accuracy. ARS DETR is a transformer-based object detector which utilizes deformable attention mechanism to address the misalignment between features and sampling points. It also introduces the aspect ratio aware circle smooth label (AR-CSL) to refine the angle label more effectively.

Quantitative evaluation results of the abovementioned algorithms are shown in Table IV. Each network's performance is assessed using two metrics. The first row shows AP50 and the second row shows AP75, distinguished by the IoU threshold used for evaluation. It can be seen that two-stage models, such as Oriented RCNN and ReDet, consistently achieve higher mAP50 and mAP75 scores than single-stage models. It is because the anchor boxes in two-stage models are better tailored to the specific characteristics of the ship targets, allowing for more accurate localization and classification. ReDet achieves the highest mAP50 and mAP75, at 81.9 and 79.3, respectively, due to its advanced rotation-equivariant convolutions and rotation-invariant backbone. These features ensure consistent feature extraction and robustness across varying object orientations, which are particularly beneficial for detecting targets like ships with high aspect ratios. Transformer-based models underperform compared to two-stage models because they require large datasets to learn complex patterns effectively. As a result, they struggle in scenarios with limited data, where two-stage models excel. Categories like CVT and MWV show lower AP due to limited instances and high intraclass diversity. Conversely, categories such as ac, CTS, and CRS achieve higher AP as they have more uniform appearance and distinctive features. In addition, most ship categories show a decrease in AP75 compared to AP50, with SUB being particularly affected. This is due to SUB's large aspect ratio, up to 8.87, making precise bounding box regression challenging.

To further elucidate the effectiveness of different algorithms on the dataset, Fig. 7 visualizes recognition results and annotations for Rotated RetinaNet, Oriented RCNN, and ReDet. Different colored boxes indicate different ship categories. Representative scenes include thick clouds, mists, sea clutter, shadows, ground facilities, combined degradation, and dense target arrangement. Overall, two-stage networks surpass single-stage ones, with ReDet showing the best

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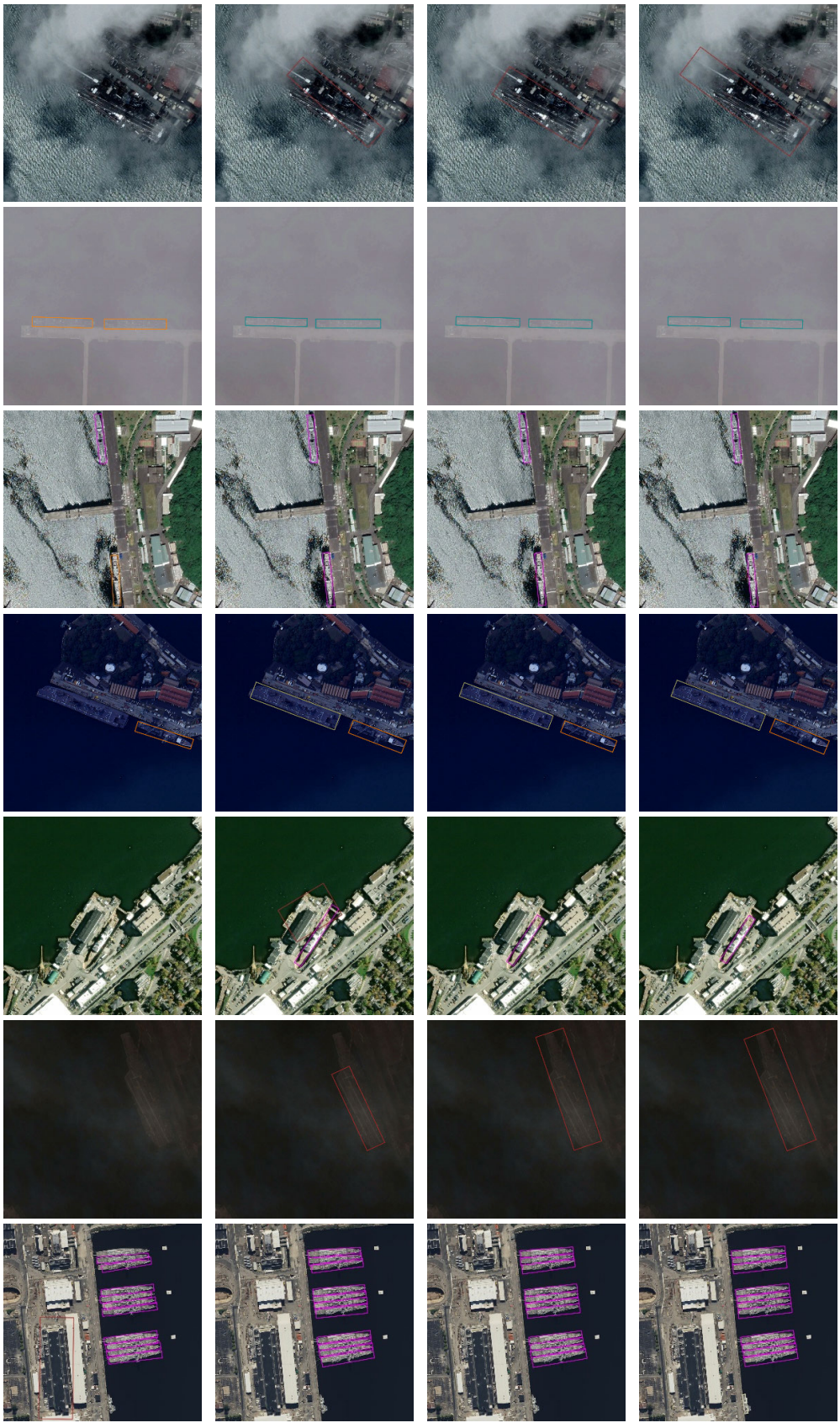


Fig. 7. Recognition results visualized by different models. (a) Rotated RetinaNet. (b) Oriented RCNN. (c) ReDet. (d) Ground truth.

performance. For thick clouds, Rotated RetinaNet misses targets due to partial occlusion, while two-stage networks struggle to accurately predict bounding boxes. Mists diminish target features, and sea clutter introduces background noise,

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resulting in misclassification for Rotated RetinaNet. For shadows, Rotated RetinaNet fails to recognize the AAS class in the image. For ground facilities, the complex textures result in Rotated RetinaNet missing targets, whereas Oriented RCNN tends to produce false alarms. For scenes combining shadows and mists, only ReDet achieves satisfactory results. For dense arrangement scenes, two-stage networks excel in accurately regressing bounding boxes, while Rotated RetinaNet tends to miss targets.

In summary, the performance differences among the three network architectures can be attributed to their structural principles. Single-stage networks, like Rotated RetinaNet, make predictions directly from the feature map, which can lead to difficulties in ambiguous environments such as partial occlusion, mist, or sea clutter. These models often miss targets or misclassify objects due to limited capacity for refining predictions. In contrast, two-stage networks, like Oriented RCNN and ReDet, first propose regions that are likely to contain objects, then refine these predictions in the second stage, offering better handling of challenging backgrounds, dense scenes, and target localization. ReDet goes a step further by extracting rotation-equivariant features, which leads to superior performance. However, they still face challenges in extreme conditions where fine-grained bounding box predictions are necessary. The results highlight that the proposed dataset presents significant challenges to current algorithms due to its inclusion of complex and varied scenarios. This complexity tests the robustness of existing recognition algorithms and provides direction for boundary exploration and further optimization, helping to improve the performance and adaptability tolerance of the final algorithms.

V. Conclusion

To advance research in ship recognition methodologies, we present the FGSRCS dataset. This comprehensive dataset includes 17 categories of large and medium-sized ships and five main complex scene elements, such as thick clouds, mists, shadows, sea clutter, and ground facilities, ensuring diversity in target types and scene elements. In terms of spatial and temporal richness of the scene, the dataset incorporates images from various satellite platforms, including Ikonos, Jilin-1, OrbView, Pleiades, and WorldView. Moreover, the dataset covers more than 280 ports and includes a substantial number of the world's military ports, with historical imagery incorporated. In addition, we conduct experiments with ten state-of-the-art recognition algorithms on the dataset, providing valuable insights into the strengths and limitations of these algorithms under challenging scenarios. Future efforts should focus on: 1) channel-spatial attention mechanisms to enhance discriminative feature learning; 2) multiscale feature fusion architectures for context-aware interpretation; and 3) targeted suppression modules for interference mitigation. The research can offer both theoretical and practical guidance for the development of future ship recognition models.

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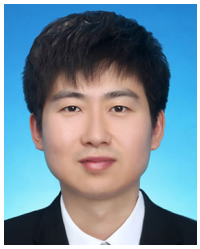
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